Received: 18 March 2022 | Revised: 18 February 2023 | Accepted: 26 February 2023

DOI: 10.1002/fld.5192

RESEARCH ARTICLE

WILEY

Airfoil inverse design based on laminar compressible adjoint lattice Boltzmann method

H. Jalali Khouzani | **R. Kamali-Moghadam**

A & S Research Institute, (Ministry of Science, Research and Technology), Tehran, Iran

Correspondence
R. Kamali-Moghadam, A & S Research Institute, Ministry of Science, Research and Technology, P.O.Box:14665-834, Tehran, Iran.
Email: rkamali@ari.ac.ir

Abstract

A new optimization techniques based on the adjoint lattice Boltzmann method is derived for airfoil inverse design in laminar compressible flows. In this study, the developed adjoint lattice Boltzmann scheme based on the circular function (CF) is extended for airfoil inverse design problems in laminar incompressible and compressible flows. New mathematical derivation based on compressible lattice Boltzmann equations (LBE) is developed which can find target shape of an airfoil with available desired pressure distribution. The adjoint lattice Boltzmann method is extended for both the incompressible and compressible flows by selecting the circular function idea for calculating the equilibrium distribution functions. So, the adjoint equation is also expanded based on CF idea for calculation of objective function gradient vector. The steepest decent technique is utilized as gradient optimizer. Also, a novel solution is presented to remove singularity problem of the adjoint boundary condition. In order to validate the developed optimization algorithm, results are presented for both incompressible and compressible inverse problem in steady and unsteady flow and accurate results are obtained.

KEYWORDS

adjoint boundary condition, adjoint method, airfoil inverse design, circular function, lattice Boltzmann method, viscous compressible flows

1 | INTRODUCTION

The computational fluid dynamics (CFD) tools give us the ability of automatic shape design for airfoil, wing or full aircraft configurations. Researchers have made great studies in this area, providing more and more accurate methods for optimizing aerodynamics based on the CFD. The adjoint method is one of the best and most well-known approaches in aerodynamic shape optimization, which has been well developed based on the Euler and Navier–Stokes equations.[1–5] Derivation of the adjoint equation depends on the flow governing equation. Thus, accuracy and performance of the adjoint method is strongly affected by the flow solvers.

In past decade, development of new flow solvers based on particle-based methods such as the lattice Boltzmann method (LBM) has been noticed by researchers in the field of aerodynamics.[6–9] The LBM is second-order Lagrangian scheme in fluid equations. This method is developed based on the Boltzmann transport equation. The simplicity of the prevailing equation compared to the Naveir-Stokes equations (NS), the simplicity of the pressure field calculation, the easier coding, inherent parallelizability, avoidance of nonlinear convective terms unlike the NS equations, and some other specific features of this method compared to the traditional CFD method have made it very popular and applicable in fluid dynamic simulations. Although the number of LBM equations depends on the used lattices and is more than the NS equations, it should be noted that the NS equations are coupled, while the LBM equations are independent and can be easily solved in parallel algorithm.[10] Development of some commercial software such as Xflow and Powerflow based on the LBM shows its importance in wide applicable flow regimes. The LBM has been applied as an alternative for the conventional CFD approaches in simulation of fluid flows. The standard LBM has been basically developed for simulation of incompressible flows.[11–14] Expanding of the standard LBM to cover the simulation of compressible flows and removing the Maxwell distribution constraint had many challenges, which makes researchers extensively study in this field.[15–22] For simulation of the compressible flowfield using the LBM, the prominent studies is performed by Qu.[17] He offered a new and alternative distribution function for the Maxwell distribution, which can satisfy all conservative constraints without any limitation in the Mach number. A major advantage of this method is the ability and high flexibility of the mathematical model to provide different lattices. This capability convinced us use the Qu's approach in the present work for simulation of compressible flows and aerodynamic shape optimization using the adjoint algorithm for the first time.

Implementation of the adjoint method in the aerodynamic shape optimization was founded by Jameson for the first time. Jameson et al.[23–26] successfully used the adjoint method based on the CFD solvers in shape optimization of airplane design. The main advantage of the adjoint method is that it can be used to obtain the gradient of the objective function independent of the number of design variables. This special advantage allows the designer to consider all surface points of an aerodynamic shape as the design variables. So, with this method a real aerodynamic shape optimization can be achieved. It is well known that the adjoint method based on the macroscopic equations, that is, NS or Euler equations has been completely developed for shape optimization. For example, Farrokhfal et al.[27,28] applied the adjoint algorithm based on the Euler equation for optimization of the rotor blade section and planform.

Recently, combination of the LBM and adjoint method has become a favorite of researchers due to their specific advantages. Very little research has been carried out in this field. Some important developments of the adjoint lattice Boltzmann method (ALBM) also Innovation and differences of this study in comparison of similar research have been prominent in the Table 1. All previous studies have been performed for optimization in incompressible flow using the standard LBM. Also, most of researches use the incompressible ALBM for flow or topological optimization. Only two works are focused on shape optimization, but they are limited to incompressible flows and the standard LBM. We recently developed a novel adjoint formulation and algorithm based on compressible LBM (Circular Function scheme) for the flow optimization in smoothed and discontinues flow.[28]

In this paper, a robust optimization algorithm is extended based on combination of the adjoint method and LBM for airfoil shape optimization for the first time. Thus, the developed code uses advantages of the LBM and also the

TABLE 1 Optimization based on adjoint method literature review.

Authors	year	Optimization type	Viscosity	Compressibility	Time dependent	Method
Jalali et al.[29]	2020	Flow	Inviscid	Incompressible/compressible	Steady/unsteady	LBM + adjoint
Cheylan et al.[30]	2019	Shape	Viscous	Incompressible	Steady	LBM + adjoint
Li et al.[31]	2018	Shape	Viscous	Incompressible	Steady	LBM + adjoint
Dugast et al.[32]	2018	Topological	Viscous	Incompressible	Steady	LBM + adjoint
Chen et al.[33]	2017	Topological	Viscous	Incompressible	Steady/unsteady	LBM + adjoint
Rokicki et al.[34]	2016	Topological	Viscous	Incompressible	Steady/unsteady	LBM + adjoint
Vergnault et al.[35]	2014	Parametric	Viscous	Incompressible	Unsteady	LBM + adjoint
Hekmat et al.[36–39]	2014	Flow	Viscous	Incompressible	Steady	LBM + adjoint
Farrokhfal et al.[27]	2013	Shape	Inviscid	Compressible	Steady	Euler + free wake + adjoint
Hu et al.[1]	2012	Shape	Inviscid	Compressible	Steady	Euler + adjoint
Tekitek et al.34	2009	Flow	Viscous	Incompressible	Unsteady	LBM + Adjoint
Nadarajah et al.[24]	2007	Shape	Viscous	Incompressible/compressible	Steady/unsteady	NS + adjoint
Present work	2022	Shape	Viscous	Incompressible/compressible	Steady/unsteady	LBM + adjoint

capabilities of the adjoint method in aerodynamic shape optimization. This algorithm requires a complex mathematical model in which the adjoint equation is derived from the lattice Boltzmann equation (LBE) and the terminal and boundary conditions are obtained accordingly. The developed ALBM is valid for both the compressible and incompressible flows. In the present study, the ALBM uses the circular function (CF) idea as the equilibrium distribution functions to apply optimization algorithm in the compressible flow. So, the adjoint equation is also expanded based on CF idea for calculation of objective function gradient vector for calculation of objective function gradient vector. The CF is used as distribution function instead of the Maxwell distribution function due to its mathematical simplicity and flexibility of new lattice definition in simulation of compressible flowfield.[17] The derivation of adjoint equations based on the CF is our major innovations, which has very complex mathematical formulations. In this research, the mathematical foundation for the adjoint compressible lattice Boltzmann method based the CF method is presented. Finally, the steepest decent technique is also utilized as gradient optimizer due to its fast convergence to reach the optimize conditions. All equation derivations of the ALBM with specific cost function are performed for the first time in this study. Also, a new approach is presented to solve singularity problem of the adjoint equation in LBE. In order to validate the new provided optimization derivation, two problems of airfoil inverse design in laminar viscous flow are presented in different conditions.

2 | DESIGN OPTIMIZATION PROBLEM USING CONTINUOUS ADJOINT METHOD

In the present paper, the cost function, I, is considered as function of the flow variables in mesoscopic scale (f) and physical location of boundary (S)

$$I = I(f, S) \tag{1}$$

so, the cost function variation can be calculated as:

$$\delta I = \frac{\partial I^T}{\partial f}\delta f + \frac{\partial I^T}{\partial s}\delta S \tag{2}$$

The flow governing equation, R, which is the LBE here, can be expressed as function of f and S within the flow domain, D,

$$R(f, S) = 0 \tag{3}$$

Thus, δf can be determined from Equation (4).

$$\delta R = \left[\frac{\partial R}{\partial f}\right]\delta f + \left[\frac{\partial R}{\partial S}\right]\delta S \tag{4}$$

By definition of the Lagrange multiplier, ψ, which is differentiable variable, the cost function variation is rewritten as:

$$\delta I = \frac{\partial I^T}{\partial f}\delta f + \frac{\partial I^T}{\partial S}\delta S - \psi^T\left(\left[\frac{\partial R}{\partial f}\right]\delta f + \left[\frac{\partial R}{\partial S}\right]\delta S\right) = \left\{\frac{\partial I^T}{\partial f} - \psi^T\left[\frac{\partial R}{\partial f}\right]\right\}\delta f + \left\{\frac{\partial I^T}{\partial S} - \psi^T\left[\frac{\partial R}{\partial S}\right]\right\}\delta S \tag{5}$$

The adjoint equation is extracted from Equation (5) by making it independence of variation of flow variables, δf.

$$\psi^T\left[\frac{\partial R}{\partial f}\right]^T = \frac{\partial I}{\partial f} \tag{6}$$

By extracting the adjoint equation, the cost function variation can be simplified as:

$$\delta I = \mathcal{G}^T\delta S \tag{7}$$

where

$$\mathcal{G}^T = \frac{\partial I^T}{\partial S} - \psi^T\left[\frac{\partial R}{\partial S}\right] \tag{8}$$

Independency of Equation (7) from variation of flow variables, δf, causes that δI can be determined by an arbitrary number of design variables without the need of additional flowfield evaluations. an improvement in Equation (7) can be made by shape change correction:

$$\delta S = -\lambda \mathcal{G} \tag{9}$$

where λ is positive, and small enough that the first variation is an accurate estimation of δI.

$$\delta I = -\lambda \mathcal{G}^T \mathcal{G} < 0 \tag{10}$$

The steepest descent can be used to minimize the gradient. To preserve the profile smoothness, the gradient is smoothed at each step. The smoothed gradient, $\overline{\mathcal{G}}$, may be calculated from a discrete approximation

$$\overline{\mathcal{G}} - \frac{\partial}{\partial \xi} \epsilon \frac{\partial}{\partial \xi} \overline{\mathcal{G}} = \mathcal{G}, \tag{11}$$

where ϵ is a smoothing parameter. Since the cost function variation usually defined as integral form in ξ-domain general one direction (1D), Equation (7) can be rewritten as:

$$\delta I = \int \mathcal{G} \delta S \, d\xi = -\int \lambda \left[\overline{\mathcal{G}} - \frac{\partial}{\partial \xi} \epsilon \frac{\partial}{\partial \xi} \overline{\mathcal{G}} \right] \overline{\mathcal{G}} \, d\xi = -\int \lambda \left[\overline{\mathcal{G}}^2 + \epsilon \left(\frac{\partial}{\partial \xi} \overline{\mathcal{G}} \right)^2 \right] d\xi < 0 \tag{12}$$

This relation guarantees that the cost function is reduced during optimization process.

3 | AIRFOIL INVERSE DESIGN USING LBE

In this section the general application of control theory for aerodynamic shape optimization is developed using the LBEs. The finite volume form of LBE for compressible flow over an airfoil may be written as

$$\frac{\partial f_{k,\upsilon}}{\partial t} + \frac{\partial F_{k,\upsilon}}{\partial x} + \frac{\partial G_{k,\upsilon}}{\partial y} - \Omega_{k,\upsilon} = 0 \; k = 1, \cdots, N \; \upsilon = 1, \ldots, 2 \tag{13}$$

where $f_{k,\upsilon}$ denotes density distribution function at time t in location of x with discrete velocity of e_k. Also, k and υ is counter of discrete velocities and energy level of the lattice in CF model, respectively. The $F_{k,\upsilon}$, $G_{k,\upsilon}$ are convective flux vectors in x and y directions, respectively. So, the lattice is defined as DMQNLO, where M, N and O indicate the problem dimension, number of discrete velocity of lattice and energy level of lattice, respectively. The BGK collision operator, $\Omega_{k,\upsilon}$, is defined as

$$\Omega_{k,\upsilon} = \frac{1}{\tau} \left(f_{k,\upsilon}^{eq} - f_{k,\upsilon} \right) \tag{14}$$

where the τ is the relaxation time and $F_{k,\upsilon}$, $G_{k,\upsilon}$ are obtained as

$$F_{k,\upsilon} = f_{k,\upsilon} e_{kx},$$

$$G_{k,\upsilon} = f_{k,\upsilon} e_{ky} \tag{15}$$

Also, $f_{k,\upsilon}^{eq}$ denotes the equilibrium distribution function which is function of macroscopic flow variables, (ρ, u, v, e). The $f_{k,\upsilon}^{eq}$ can be analytically obtained from the CF idea which is explained in details in Reference [17].The macroscopic properties of flow can be evaluated from mesoscopic states as

$$\rho = \sum_{k=1}^{N} \sum_{\upsilon=1}^{2} f_{k,\upsilon}$$
$$u = \frac{1}{\rho} \sum_{k=1}^{N} \sum_{\upsilon=1}^{2} f_{k,\upsilon} e_{kx}$$

$$v = \frac{1}{\rho}\sum_{k=1}^{N}\sum_{\upsilon=1}^{2} f_{k,\upsilon} e_{ky}$$
$$V = \frac{1}{\rho}\sum_{k=1}^{N}\sum_{\upsilon=1}^{2} f_{k,\upsilon} e_{k}$$
$$E = \frac{1}{\rho}\sum_{k=1}^{N}\left[\sum_{\upsilon=1}^{2}\frac{f_{k,\upsilon}\boldsymbol{e}_k^2}{2} + f_{k,2}\right]$$
$$e = E - \frac{V^2}{2} \tag{16}$$

where ρ, u, v, E and e are density, velocity components in x and y directions, total energy and static energy, respectively. $V = \sqrt{u^2+v^2}$ denotes the total velocity and $e_k = \sqrt{e_{kx}^2 + e_{ky}^2}$ is the total discrete lattice velocity.

The LBE (Equation 13) can be rewritten in general curvilinear coordinate as

$$J\frac{\partial f_{k,\upsilon}}{\partial t} + \frac{\partial F_{k,\upsilon}}{\partial \xi} + \frac{\partial G_{k,\upsilon}}{\partial \eta} - \Omega_{k,\upsilon} = 0 \;\; in\; D, k = 1, \cdots, N \;\; \upsilon = 1, \cdots, 2 \tag{17}$$

where

$$\Omega_{k,\upsilon} = J\frac{1}{\tau}\left(f_{k,\upsilon}^{eq} - f_{k,\upsilon}\right), F_{k,\upsilon} = J\left(f_{k,\upsilon}e_{k\xi}\right), G_{k,\upsilon} = J\left(f_{k,\upsilon}e_{k\eta}\right) \tag{18}$$

and (ξ, η) represents the computational coordinate system and the Jacobean is introduced as

$$J = \det(K) = \frac{\partial x}{\partial \xi}\frac{\partial y}{\partial \eta} - \frac{\partial x}{\partial \eta}\frac{\partial y}{\partial \xi}. \tag{19}$$

In this paper, the unsteady inverse problem is defined for minimizing the cost function of pressure difference in least square form,

$$I = \frac{1}{2}\int_0^T \int_C (p-p_d)^2 ds dt = \frac{1}{2}\int_0^T \int_C (p-p_d)^2 \left(\frac{ds}{d\xi}\right) d\xi dt \tag{20}$$

where p_d is the desired pressure. The design problem is now treated as a control problem where the control function is the airfoil shape for minimizing the cost function, I, with the constraints defined by the flow governing Equation (17). Every shape variation will result in pressure variation, δp. Thus, the variation in the cost function can be expressed as.

$$\delta I = \int_0^T \int_C (p-p_d)\,\delta p \left(\frac{ds}{d\xi}\right) d\xi dt + \frac{1}{2}\int_0^T \int_C (p-p_d)^2\, \delta\left(\frac{ds}{d\xi}\right) d\xi dt. \tag{21}$$

By multiplying a vector, $\psi_{k,\upsilon}$ in the governing Equation (17) and integrating over the time, domain, discrete velocities and energy levels, the following equation can be obtained:

$$\sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}\left(J\frac{\partial f_{k,\upsilon}}{\partial t} + Je_{k\xi}\frac{\partial f_{k,\upsilon}}{\partial \xi} + Je_{k\eta}\frac{\partial f_{k,\upsilon}}{\partial \eta} - J\Omega_{k,\upsilon}\right) d\xi d\eta dt = 0 \tag{22}$$

If $\psi_{k,\upsilon}$ is differentiable, variation of Equation (22) can be added to Equation (21) as a constraint.

$$\begin{aligned}\delta I = &\int_0^T \int_C (p-p_d)\,\delta p \left(\frac{ds}{d\xi}\right) d\xi dt + \frac{1}{2}\int_0^T \int_C (p-p_d)^2\, \delta\left(\frac{ds}{d\xi}\right) d\xi dt\\ &+ \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}\delta\left(J\frac{\partial f_{k,\upsilon}}{\partial t} + Je_{k\xi}\frac{\partial f_{k,\upsilon}}{\partial \xi} + Je_{k\eta}\frac{\partial f_{k,\upsilon}}{\partial \eta} - J\Omega_{k,\upsilon}\right) d\xi d\eta dt,\end{aligned} \tag{23}$$

Thus, the variation of the cost function may be rewritten.

$$\delta I = \int_0^T \int_C (p - p_d)\, \delta p \left(\frac{ds}{d\xi}\right) d\xi dt + \frac{1}{2}\int_0^T \int_C (p - p_d)^2\, \delta \left(\frac{ds}{d\xi}\right) d\xi dt$$
$$+ \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}\left(\frac{\partial f_{k,\upsilon}}{\partial t}\delta(J) + \frac{\partial f_{k,\upsilon}}{\partial \xi}\delta\left(Je_{k\xi}\right) + \frac{\partial f_{k,\upsilon}}{\partial \eta}\delta\left(Je_{k\eta}\right) - \Omega_{k,\upsilon}\delta(J)\right) d\xi d\eta dt$$
$$+ \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \left(\frac{\partial\left(J\psi_{k,\upsilon}\delta f_{k,\upsilon}\right)}{\partial t} + \frac{\partial\left(J\psi_{k,\upsilon}e_{k\xi}\delta f_{k,\upsilon}\right)}{\partial \xi} + \frac{\partial\left(J\psi_{k,\upsilon}e_{k\eta}\delta f_{k,\upsilon}\right)}{\partial \eta}\right) d\xi d\eta dt$$
$$- \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \left(\frac{\partial\left(J\psi_{k,\upsilon}\right)}{\partial t} + \frac{\partial\left(J\psi_{k,\upsilon}e_{k\xi}\right)}{\partial \xi} + \frac{\partial\left(J\psi_{k,\upsilon}e_{k\eta}\right)}{\partial \eta} + \sum_{j=1}^{N}\sum_{\upsilon=1}^{M}\psi_{j,\upsilon}\frac{\partial \Omega_{j,\upsilon}}{\partial f_{k,\upsilon}}\delta f_{k,\upsilon}\right) d\xi d\eta dt \tag{24}$$

According to Gauss theorem, the domain integral can be converted to boundary integral; therefore no boundary integrals appear in the η-direction because the O-type mesh and periodic boundary in the ξ-coordinate are used around the airfoil, thereby canceling the η boundary integrals (see Figure 1). So, the fourth term of Equation (24) can be written as.

$$\sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \left(\frac{\partial\left(J\psi_{k,\upsilon}e_{k\xi}\delta f_{k,\upsilon}\right)}{\partial \xi} + \frac{\partial\left(J\psi_{k,\upsilon}e_{k\eta}\delta f_{k,\upsilon}\right)}{\partial \eta}\right)$$
$$d\xi d\eta dt = \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_B \left(n_1\psi_{k,\upsilon}e_{k\xi} + n_2\psi_{k,\upsilon}e_{k\eta}\right) J\delta f_{k,\upsilon}\, d\xi dt$$
$$+ \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_C \left(n_1\psi_{k,\upsilon}e_{k\xi} + n_2\psi_{k,\upsilon}e_{k\eta}\right) J\delta f_{k,\upsilon}\, d\xi dt \tag{25}$$

where B denotes the farfield boundary, C is the airfoil surface boundary and $\psi_{k,\upsilon}$ is zero at farfield. Then, the variation of the cost function is rewritten as.

$$\delta I = \int_0^T \int_C (p - p_d)\, \delta p \left(\frac{ds}{d\xi}\right) d\xi dt + \frac{1}{2}\int_0^T \int_C (p - p_d)^2\, \delta \left(\frac{ds}{d\xi}\right) d\xi dt$$
$$+ \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}\left(\frac{\partial f_{k,\upsilon}}{\partial t}\delta(J) + \frac{\partial f_{k,\upsilon}}{\partial \xi}\delta\left(Je_{k\xi}\right) + \frac{\partial f_{k,\upsilon}}{\partial \eta}\delta\left(Je_{k\eta}\right) - \Omega_{k,\upsilon}\delta(J)\right) d\xi d\eta dt$$
$$+ \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \frac{\partial\left(J\psi_{k,\upsilon}\delta f_{k,\upsilon}\right)}{\partial t} d\xi d\eta dt - \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_C \left(J\psi_{k,\upsilon}e_{k\eta}\right)\delta f_{k,\upsilon}\, d\xi dt$$
$$- \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \left(\frac{\partial \psi_{k,\upsilon}}{\partial t} + e_{k\xi}\frac{\partial \psi_{k,\upsilon}}{\partial \xi} + e_{k\eta}\frac{\partial \psi_{k,\upsilon}}{\partial \eta} + \sum_{j=1}^{N}\sum_{\upsilon=1}^{M}\psi_{j,\upsilon}\frac{\partial \Omega_{j,\upsilon}}{\partial f_{k,\upsilon}}\delta f_{k,\upsilon}\right) J\,\delta f_{k,\upsilon}\, d\xi d\eta dt \tag{26}$$

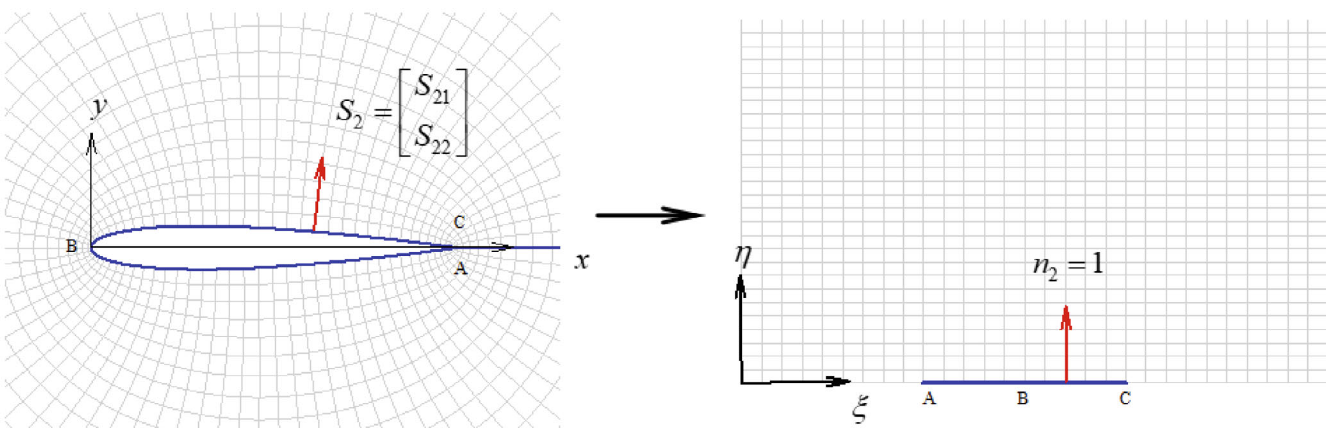


FIGURE 1 Mapping the O-type grid physical domain to the computational domain for an airfoil. [Colour figure can be viewed at wileyonlinelibrary.com]

In order to independency of Equation (26) from flow variations, the last term must be zero.

$$\frac{\partial \psi_{k,\upsilon}}{\partial t} + e_{k\xi}\frac{\partial \psi_{k,\upsilon}}{\partial \xi} + e_{k\eta}\frac{\partial \psi_{k,\upsilon}}{\partial \eta} + \sum_{j=1}^{N}\sum_{\upsilon=1}^{M}\psi_{j,\upsilon}\frac{\partial \Omega_{j,\upsilon}}{\partial f_{k,\upsilon}}\delta f_{k,\upsilon} = 0 \tag{27}$$

Equation (27) is the adjoint lattice Boltzmann equation (ALBE). $\psi_{k,\upsilon}$ is obtained from solution of the adjoint equation. The third term of equation (27), called BGK adjoint collision operator, can be written as

$$\Theta_{k,\upsilon} = \sum_{j=1}^{N}\sum_{\upsilon=1}^{M}\psi_{j,\upsilon}\frac{\partial \Omega_{j,\upsilon}}{\partial f_{k,\upsilon}} = \frac{1}{\tau}\left(\psi_{k,\upsilon}^{eq} - \psi_{k,\upsilon}\right) \tag{28}$$

where $\psi_{k,\upsilon}^{eq}$ is the adjoint equilibrium distribution function.

$$\psi_{k,\upsilon}^{eq} = \sum_{j=1}^{N}\sum_{\upsilon=1}^{M}\frac{\partial f_{k,\upsilon}^{eq}}{\partial f_{k,\upsilon}}\psi_{j,\upsilon}. \tag{29}$$

The adjoint equilibrium distribution function is derived from partial derivatives of the CF formulas.

$$\frac{\partial f_{\alpha,\upsilon}^{eq}}{\partial f_{\alpha,\upsilon}} = \frac{\partial f_{\alpha,\upsilon}^{eq}}{\partial \rho}\frac{\partial \rho}{\partial f_{\alpha,\upsilon}} + \frac{\partial f_{\alpha,\upsilon}^{eq}}{\partial u}\frac{\partial u}{\partial f_{\alpha,\upsilon}} + \frac{\partial f_{\alpha,\upsilon}^{eq}}{\partial e}\frac{\partial e}{\partial f_{\alpha,\upsilon}}, \tag{30}$$

The derivatives of the macroscopic variables in the CF formulas can be analytically obtained.

$$\begin{aligned}
\frac{\partial \rho}{\partial f_{k,\upsilon}} &= 1\\
\frac{\partial u}{\partial f_{k,\upsilon}} &= \frac{1}{\rho}\left(e_{kx} - u\right)\\
\frac{\partial v}{\partial f_{k,\upsilon}} &= \frac{1}{\rho}\left(e_{ky} - v\right)\\
\frac{\partial V}{\partial f_{k,\upsilon}} &= \frac{1}{\rho}\left(e_{k} - V\right)\\
\frac{\partial e}{\partial f_{k,\upsilon}} &= \frac{1}{2\rho}\left(-2E + (V - e_k)^2 + V^2 + 2(\upsilon - 1)\right)
\end{aligned} \tag{31}$$

Using these relations, $\psi_{k,\upsilon}^{eq}$ can be calculated by some mathematical operations. Equation (31) is presented first with this work for shape optimization in the compressible flow based on LBM.

3.1 | Adjoint terminal condition

Since the adjoint equation (Equation 27) has a temporary term, we must be determined initial condition for solution. The adjoint equation is solved backward in time, so, the temporary condition or terminal condition can be defined using the temporary term of this equation:

$$\sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T\int_D \frac{\partial\left(J\psi_{k,\upsilon}\delta f_{k,\upsilon}\right)}{\partial t}\,d\xi d\eta dt = \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_D \left\{\psi_{k,\upsilon}(T)\delta f_{k,\upsilon}(T) - \psi_{k,\upsilon}^{eq}\delta f_{k,\upsilon}(0)\right\} J\, d\xi d\eta = 0, \tag{32}$$

then

$$\psi_{k,\upsilon}(T) = 0, \tag{33}$$

3.2 | Adjoint viscous boundary condition

The boundary condition of ALBE can be extracted with canceling the gain of flow variation in the boundary integral of Equation (27).

$$\int_0^T \int_C (p - p_d)\, \delta p \left(\frac{ds}{d\xi}\right) d\xi dt - \sum_{k=1}^{N}\sum_{\upsilon=1}^{M} \int_0^T \int_C \left(J\psi_{k,\upsilon} e_{k\eta}\right) \delta f_{k,\upsilon}\, d\xi dt = 0, \tag{34}$$

The pressure can be written as function of distribution function, $f_{k,\upsilon}$, computed by the CF scheme, using the state equation, $p = (\gamma - 1)\rho e$, and definition of the macroscopic variables (Equation (16)), Therefore,

$$\frac{\partial p}{\partial f_{k,\upsilon}} = \frac{(\gamma - 1)}{2}\left((V - e_k)^2 + 2(\upsilon - 1)\right) \tag{35}$$

By considering $V = 0$ on the wall, the relation between δf and δp is

$$\delta p = \frac{\partial p}{\partial f_{k,\upsilon}} \delta f_{k,\upsilon} = \frac{(\gamma - 1)}{2} \sum_{k=1}^{N}\sum_{\upsilon=1}^{M} \left(e_k^2 + 2(\upsilon - 1)\right) \delta f_{k,\upsilon} \text{ on C.} \tag{36}$$

By substituting Equation (36) to Equation (34), the adjoint boundary condition is calculated as.

$$\psi_{k,\upsilon} = \frac{1}{2}\left(\frac{(\gamma - 1)(p - p_d)\left(\frac{ds}{d\xi}\right)}{Je_{k\eta}}\right)\left(e_k^2 + 2(\upsilon - 1)\right) \text{ on C.} \tag{37}$$

The singularity problem appears when the static particle of lattice is zero ($e_{k\eta} = 0$). The simple and novel method to avoiding such singularity is to eliminate the static particle from lattice, because the CF scheme is very flexible to creation of new lattice. Thus, a new D2Q12L2 lattice without zero velocity particle is proposed for the first time to overcome the singularity (Figure 2).

Although it seems simple way, it is not easy to apply in LBM. In fact, the D2Q12L2 lattice generation without static particle is the new challenge. The lattice introduction based on C.F. idea is very difficult process. For this purpose, the third interpolation Lagrangian polynomial must be selected to help in constructing assigning function (more details are in Reference [42]). The actual form of polynomials depends on the configuration of the lattice in the velocity space. It is needed to construct a third order 2D polynomial. In the 2D space ((x–y) plan), after some mathematical efforts, the 12-node interpolation polynomial can be derived using some mathematical tools (like Maple software), the base function and then the equilibrium distribution functions based on CF scheme can be obtained for this new lattice. These obtained

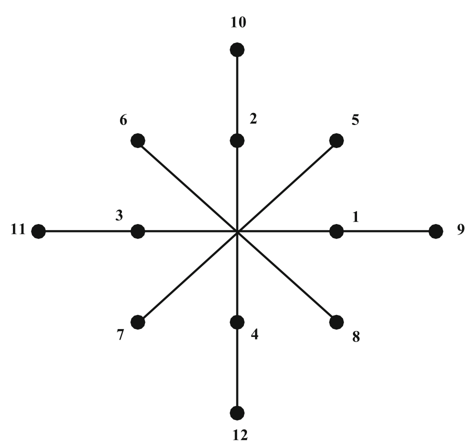


FIGURE 2 New proposed D2Q12L2 lattice.

relations for base and distributions functions are very large expressions and could not be written in the present article for summarizing.

It should be noted that both flow and adjoint equilibrium distribution function can be analytically calculated. The computed $f_{k,\upsilon}^{eq}$ causes that Equation (13) recovers the compressible NS equations. The values of discrete velocity vectors in proposed lattice are expressed as.

$$e_k = \begin{cases} cyc:(\pm 1, 0) & i = 1,2,3,4 \\ cyc:(\pm 1, \pm 1) & i = 5,6,7,8 \\ cyc:(\pm 2, 0) & i = 9,10,11,12 \end{cases} \tag{38}$$

By removing the singularity, the cost function variation may finally expressed as.

$$\begin{aligned} \delta I = &\frac{1}{2}\int_0^T \int_C (p-p_d)^2 \, \delta\left(\frac{ds}{d\xi}\right) d\xi dt + \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}\left(\frac{\partial f_{k,\upsilon}}{\partial t}\delta J + \frac{\partial f_{k,\upsilon}}{\partial \xi}\delta\left(Je_{k\xi}\right) + \frac{\partial f_{k,\upsilon}}{\partial \eta}\delta\left(Je_{k\eta}\right) - \Omega_{k,\upsilon}\delta J\right) d\xi d\eta dt \\ &- \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}^{eq}\frac{\partial f_{k,\upsilon}^{eq}}{\partial f_{k,\upsilon}} J \, d\xi d\eta dt \end{aligned} \tag{39}$$

Since the last term do not have any variation of metrics, it is vanished. Then δI can rewritten as.

$$\begin{aligned} \delta I = &\frac{1}{2}\int_0^T \int_C (p-p_d)^2 \, \delta\left(\frac{ds}{d\xi}\right) d\xi dt + \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_C \psi_{k,\upsilon}\left(\frac{\partial f_{k,\upsilon}}{\partial t}(\delta J)\right) d\xi d\eta dt \\ &+ \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}\left(\frac{\partial f_{k,\upsilon}}{\partial \xi}\delta\left(Je_{k\xi}\right)\right) d\xi d\eta dt + \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}\left(\frac{\partial f_{k,\upsilon}}{\partial \eta}\delta\left(Je_{k\eta}\right)\right) d\xi d\eta dt \\ &- \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_D \psi_{k,\upsilon}\left(\Omega_{k,\upsilon}\delta J\right) d\xi d\eta dt \end{aligned} \tag{40}$$

Now it is possible to project volume integrals terms to surface integrals of airfoil. If we use the Jameson's grid modification,40 we have

$$\begin{aligned} x^{\text{new}} &= x^{\text{old}} + \mathcal{R}\left(x_s^{\text{new}} - x_s^{\text{old}}\right) \\ y^{\text{new}} &= y^{\text{old}} + \mathcal{R}\left(y_s^{\text{new}} - y_s^{\text{old}}\right) \end{aligned} \tag{41}$$

where x and y represent the domain grid points and, x_s and y_s, denote the surface grid points. $\mathcal{R}$ is also calculated as function of η.

$$\mathcal{R} = 1 - \left(3 - 2\nu_j\right)\nu_j^2 \tag{42}$$

where

$$\mathrm{N}_j = \frac{\sum_{l=2}^{j}\sqrt{\left(x_{i,l} - x_{i,l-1}\right)^2 + \left(y_{i,l} - y_{i,l-1}\right)^2}}{\sum_{l=2}^{j\max}\sqrt{\left(x_{i,l} - x_{i,l-1}\right)^2 + \left(y_{i,l} - y_{i,l-1}\right)^2}}, \quad j \geq 2 \tag{43}$$

The required variations in the metric terms can then be obtained in terms of surface perturbations since it follows that

$$\begin{aligned} \delta x &= \mathcal{R}\,\delta x_s \\ \delta y &= \mathcal{R}\,\delta y_s. \end{aligned} \tag{44}$$

and

$$[K] = \mathcal{R}\ [K_s] \tag{45}$$

where K_s is the transformation matrix K defined at the surface. Now all of metrics variations in the cost function gradient can be evaluated along surface direction. So, it is convenient to rewrite (40) as:

$$\begin{aligned}\delta I = &\frac{1}{2}\int_0^T \int_C \left\{ \delta\left(\frac{\partial x}{\partial \xi}\right)_s \mathcal{M}_{11} + \delta\left(\frac{\partial y}{\partial \xi}\right)_s \mathcal{M}_{12} + \delta(x)_s\mathcal{M}_{13} + \delta(y)_s\mathcal{M}_{14} \right\} d\xi dt \\ &+ \sum_{k=1}^{N}\sum_{\upsilon=1}^{M}\int_0^T \int_C \left\{ \delta\left(\frac{\partial x}{\partial \xi}\right)_s [\mathcal{M}_{21}+\mathcal{M}_{31}+\mathcal{M}_{41}-\mathcal{M}_{51}] + \delta\left(\frac{\partial y}{\partial \xi}\right)_s [\mathcal{M}_{22}+\mathcal{M}_{32}+\mathcal{M}_{42}-\mathcal{M}_{52}] \right. \\ &\left. + \delta(x)_s [\mathcal{M}_{23}+\mathcal{M}_{33}+\mathcal{M}_{43}-\mathcal{M}_{53}] + \delta(y)_s [\mathcal{M}_{24}+\mathcal{M}_{34}+\mathcal{M}_{44}-\mathcal{M}_{54}] \right\} d\xi dt\end{aligned} \tag{46}$$

Finally, if the matrix $\mathcal{M}_{ij}$ is determined the cost function gradient $\mathcal{G}$ is obtained. For example the metric variation $\delta\left(\frac{ds}{d\xi}\right)$ can be analytically calculated as:

$$\delta\left(\frac{ds}{d\xi}\right) = -\frac{\left(\frac{\partial x}{\partial \xi}\right)\mathcal{R}}{\sqrt{\left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2}}\, \delta\left(\frac{\partial x}{\partial \xi}\right)_s - \frac{\left(\frac{\partial y}{\partial \xi}\right)\mathcal{R}}{\sqrt{\left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2}}\, \delta\left(\frac{\partial y}{\partial \xi}\right)_s \tag{47}$$

so, we have

$$\begin{aligned}\mathcal{M}_{11} &= -\left(\frac{\left(\frac{\partial x}{\partial \xi}\right)\mathcal{R}}{\sqrt{\left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2}}\right)(p-p_d)^2 \\ \mathcal{M}_{12} &= -\left(\frac{\left(\frac{\partial y}{\partial \xi}\right)\mathcal{R}}{\sqrt{\left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2}}\right)(p-p_d)^2 \\ \mathcal{M}_{13} &= \mathcal{M}_{14} = 0.\end{aligned} \tag{48}$$

Other elements of matrix can be calculated with the same roles.

4 | IMPLEMENTATION OF PROPOSED SHAPE OPTIMIZATION METHOD

Procedures of implementation of proposed optimization method can be summarized as follow.

Solve the LBE (Equation 17) for finding $f_{k,\upsilon}, \rho, u, v$, and e.

1. Solve the adjoint equation (Equation 27) for finding $\psi_{k,\upsilon}$ subject to the terminal condition (Equation 33) and boundary condition (Equation 37).
2. Smooth the $\psi_{k,\upsilon}$ if necessary (by using Sobolev inner product).
3. Calculate matrix of $\mathcal{M}_{ij}$ according to metrics variations.
4. Calculate cost function gradient, $\mathcal{G}$, using Equation (46).
5. Smooth the $\mathcal{G}$ by Sobolev inner product to determined $\overline{\mathcal{G}}$ using Equation (11).
6. Update the design variables (x_s and y_s) based on the steepest decent method.

$$\delta x_s = -\lambda\overline{\mathcal{G}}_x \delta y_s = -\lambda\overline{\mathcal{G}}_y. \tag{49}$$

7. Return to step 1 until the convergence criterion is satisfied.

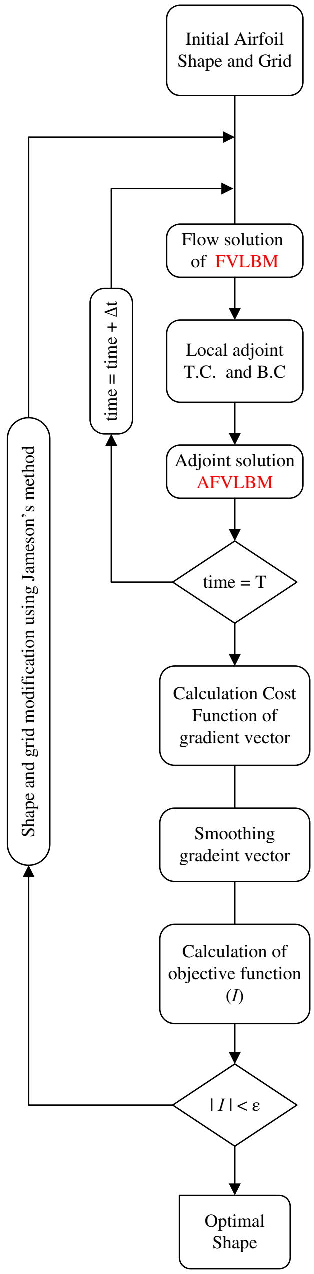


FIGURE 3 Proposed unsteady LTALBM shape optimization algorithm. [Colour figure can be viewed at wileyonlinelibrary.com]

Flow chart of the proposed optimization algorithm is shown in Figure 3.This algorithm is local in time adjoint LBM (LTALBM), that is, the adjoint and flow variables are obtained in one time step for decreasing computer RAM implementation.

5 | NUMERICAL SOLUTION OF FLOW AND ADJOINT EQUATIONS

The finite volume method (FVM) has been used here for the numerical solution of flow and adjoint equations. To discretize the 2D LBE for flow simulation using the cell centered finite volume method, the conservative form, that is, Equation (17) should be used. The finite volume discrete form of the LBE can be written as:

$$f^{n+1}_{k,\upsilon,i,j} = f^{n}_{k,\upsilon,i,j} - \frac{\Delta t}{J_{i,j}} \left\{ \left(F^{n}_{k,\upsilon,i+\frac{1}{2},j} - F^{n}_{k,\upsilon,i-\frac{1}{2},j} \right) + \left(G^{n}_{k,\upsilon,i,j+\frac{1}{2}} - G^{n}_{k,\upsilon,i,j-\frac{1}{2}} \right) + \Omega_{k,\upsilon,i,j} \right\} \tag{50}$$

In Equation (50), n is the old time, i, j are the cell centers, $F_{k,\upsilon,i\pm\frac{1}{2},j}$ and $G_{k,\upsilon,i,j\pm\frac{1}{2}}$ are the numerical fluxes. Numerical fluxes can be calculated by the Riemann solver and interpolation methods. To capture the shock waves in numerical simulations of the compressible flow, the numerical dissipation or artificial viscosity is applied to the solver. For spatial discretization of the numerical fluxes, the third order of MUSCL method is used with a smooth limiter to interpolate the fluxes on both sides of the cell surface. For example, F flux can be discretized as:

$$F^{n}_{k,\upsilon,i+\frac{1}{2},j} = \begin{cases} (f_{\mathrm{L}})_{k,\upsilon,i+\frac{1}{2},j} J \text{ if } \boldsymbol{e}_{k\xi} \geq 0 \\ (f_{\mathrm{R}})_{k,\upsilon,i+\frac{1}{2},j} J \text{ if } \boldsymbol{e}_{k\xi} \leq 0 \end{cases} \tag{51}$$

The left (f_{L}) and right (f_{R}) distribution functions of the surface can be written as:

$$\begin{cases} (f_L)_{k,\upsilon,i+\frac{1}{2},j} = f_{k,\upsilon,i,j} + \left\{ \frac{s}{4} \left[(1-Ks)\Delta_- + (1+Ks)\Delta_+ \right] \right\}_i \\ (f_R)_{k,\upsilon,i+\frac{1}{2},j} = f_{k,\upsilon,i+1,j} - \left\{ \frac{s}{4} \left[(1-Ks)\Delta_+ + (1+Ks)\Delta_- \right] \right\}_{i+1} \end{cases} \tag{52}$$

where K determines accuracy of the interpolation. For example, $K = 1/3$, 0, 1 gives third, upwind second and central second order of interpolations, respectively. The Van Albada limiter, s, is also defined as:

$$s = \frac{2\Delta_+\Delta_+ + \varepsilon^2}{\Delta_+^2 + \Delta_-^2 + \varepsilon} \tag{53}$$

where ε has a very small value about 10^{-6} and the Δ operator is expressed as:

$$(\Delta_+)_i = f_{k,\upsilon,i+1,j} - f_{k,\upsilon,i,j} \; (\Delta_-)_i = f_{k,\upsilon,i,j} - f_{k,\upsilon,i-1,j} \tag{54}$$

Other numerical fluxes are also calculated in the same way.

In order to discretize the ALBE for calculation of the objective function gradients which are independent to number of the design variables, the same method applied to the LBE is used. Therefore, the ALBE in conservative form in computational domain in 2D, that is, Equation (27) can be considered. Then the finite volume discrete form of the ALBE can be written as:

$$\psi^{n-1}_{k,\upsilon,i,j} = \psi^{n}_{k,\upsilon,i,j} - \frac{\Delta t}{J_{i,j}} \left\{ \left(\mathcal{F}^{n}_{k,\upsilon,i+\frac{1}{2},j} - \mathcal{F}^{n}_{k,\upsilon,i-\frac{1}{2},j} \right) + \left(\mathcal{R}^{n}_{k,\upsilon,i,j+\frac{1}{2}} - \mathcal{R}^{n}_{k,\upsilon,i,j-\frac{1}{2}} \right) + \Theta_{k,\upsilon,i,j} \right\} \tag{55}$$

where $\mathcal{F}_{k,\upsilon,i\pm\frac{1}{2},j}$ and $\mathcal{R}_{k,\upsilon,i,j\pm\frac{1}{2}}$ are the adjoint numerical fluxes. The similar way is adopted to solve Equation (55) as well as flow equation but backward in time.

6 | LAMINAR FLOW SIMULATION RESULTS

To validate the flow solver, firstly, flows around the NACA0012 airfoil are simulated in both incompressible and compressible laminar conditions with the new D2Q12L2 lattice. The captured results are compared with those of the known valid *Xfoil* code.[41] Incompressible Flow simulation is performed at conditions of $M = 0.25$, Re = 500, $\alpha = 3°$. Figure 4 indicates that the captured results are excellently well-matched with the validated Xfoil code for both normal and shear loads.

As another validation, compressible flow at conditions of $M = 0.5$, Re = 5000, $\alpha = 0°$ is also simulated and flow contours and surface coefficient distribution are compared with those of the Xfoil code in Figure 4. Although reasonable

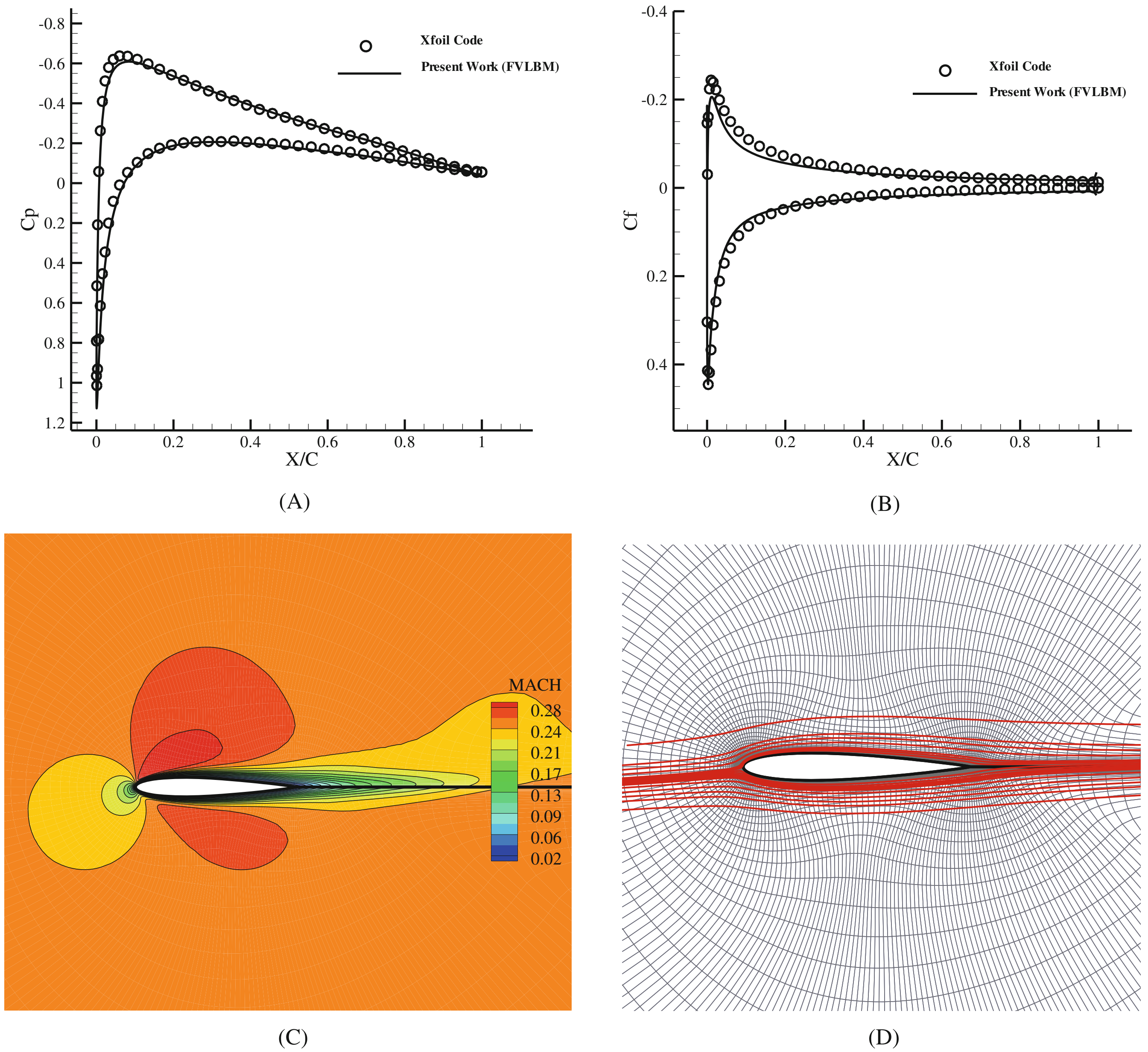


FIGURE 4 Incompressible laminar flow simulation comparison around the NACA0012, $M = 0.25$, Re $= 500$, $\alpha = 3°$, (A) Pressure coefficient distribution, (B) Skin friction coefficient distribution, (C) Mach No. contour, (D) Grid and stream lines. [Colour figure can be viewed at wileyonlinelibrary.com]

agreement has been obtained, there is a little different between *Cp* distributions at separation bubble in the trailing edge region (Figure 5A) due to none-zero bulk viscosity in the FVLBM method. In the BGK collision model has only one parameter, τ, to adjust all the three transport coefficients, that is, viscosity (μ), bulk viscosity (μ_b), and heat conductivity (k). Therefore, the Pr number must be equal to 1 in the FVLBM.[17,42]

7 | DESIGN OPTIMIZATION RESULTS

In order to validate the developed optimization algorithm, two test case problem of airfoil inverse design are studied. The first problem is the conversion of the NACA2410 to NACA0012 airfoil at the unsteady laminar incompressible flow. The second problem is the conversion of ONERA-D airfoil (airfoil of ONERA M6 wing) to transonic CAST10-2

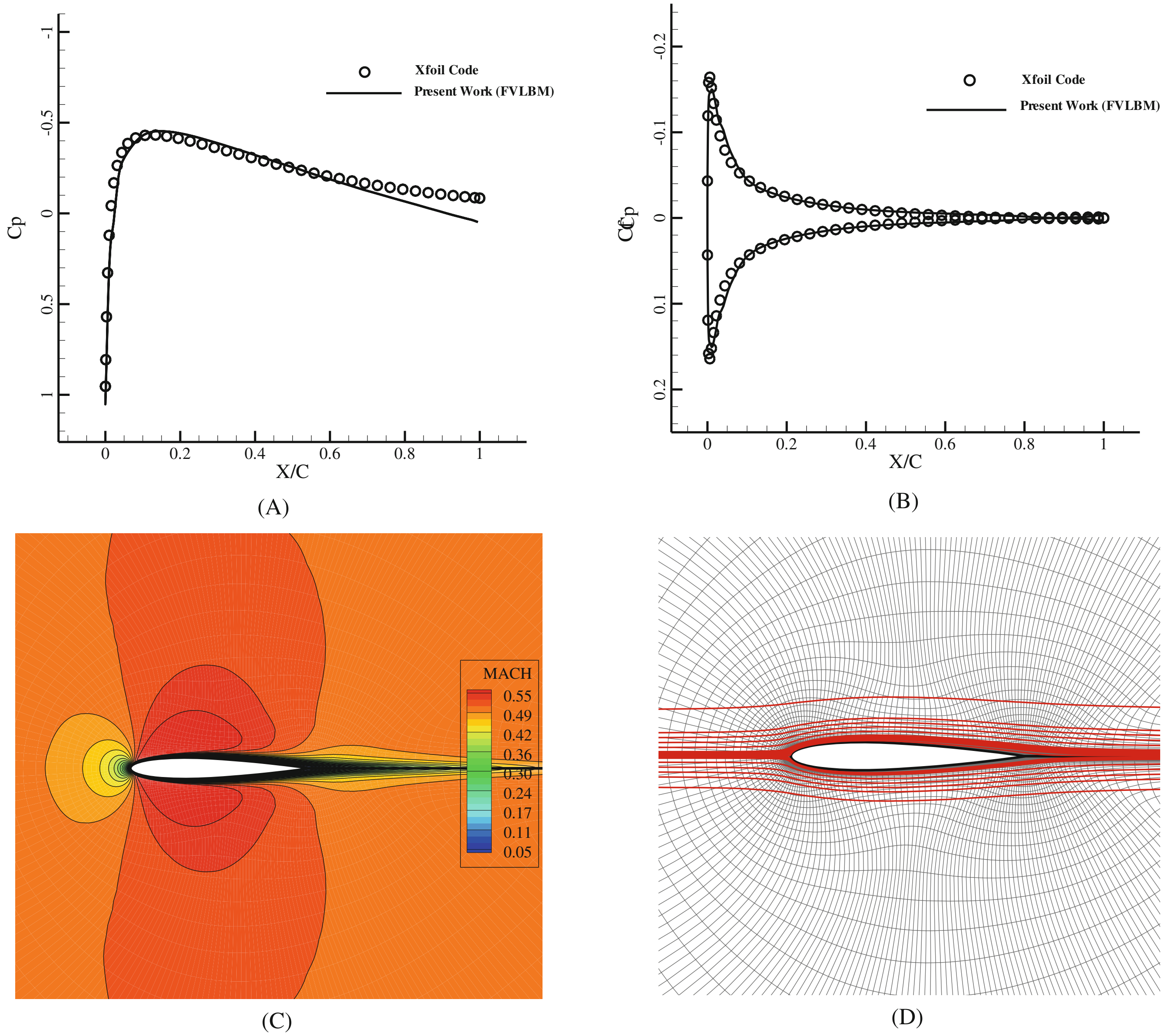


FIGURE 5 Compressible laminar flow simulation comparison around the NACA0012, $M = 0.5$, $\text{Re} = 5000$, $\alpha = 0^{\circ}$, (A) Pressure coefficient distribution, (B) Skin friction coefficient distribution, (C) Mach No. contour, and (D) Grid and stream lines. [Colour figure can be viewed at wileyonlinelibrary.com]

at the viscous compressible steady flow. In those validation cases, we tend to finding the shape of a airfoil that we have it is *Cp* distribution in certain flow free stream conditions. This process start via other airfoil and shape is changed during optimization process, finally the shape of target airfoil can be find out. Therefore, by evaluation the results of these two problems, capability of the proposed optimization algorithm can be examined for different flow conditions.

7.1 | Case I: Unsteady incompressible laminar flow ($M_{\infty} = 0.25, \text{Re}_{\infty} = 500, \alpha = 3.5°$)

In the test case, the NACA2410 airfoil is unsteadily modified to achieve the pressure distribution of the NACA0012 airfoil at final time 1.5. The optimization problem is performed at Mach number of 0.25, Reynolds number of 500 and 3.5° angle of attack. The total number of surface point design variables is 81 nodes. The cost function gradient versus design variables is shown in Figure 6 for the first design cycle. For the validation of derivatives, these values are compared with

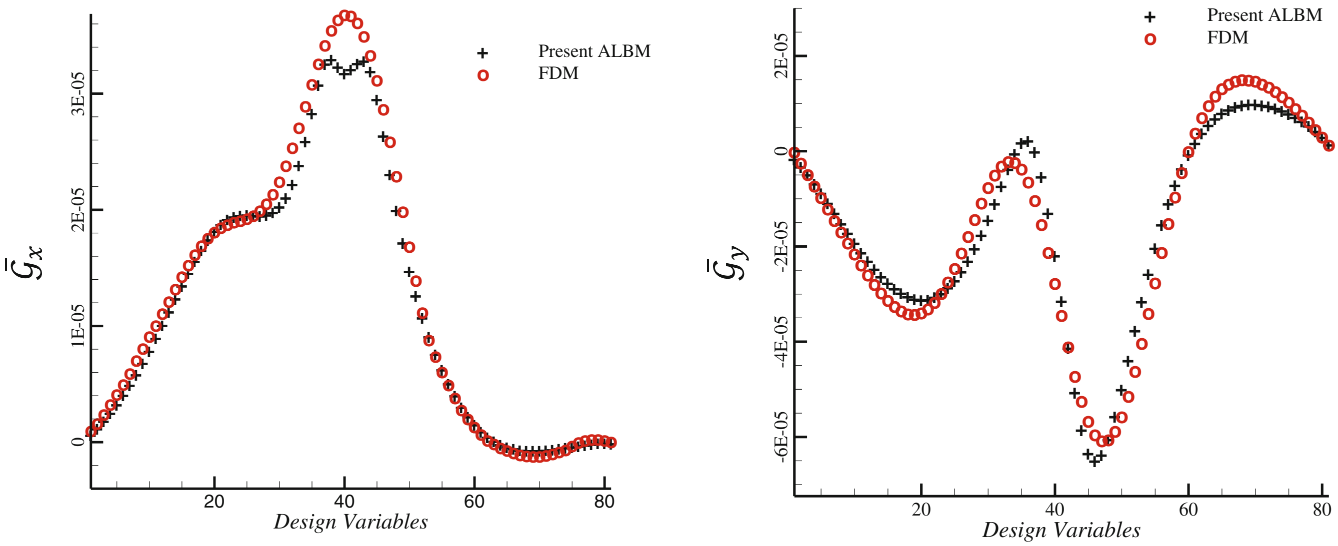


FIGURE 6 Distribution of smoothed gradients of the cost function $\left(\overline{\mathcal{G}}_x, \overline{\mathcal{G}}_y\right)$, comparison of present ALBM and FDM gradient in first design cycle. [Colour figure can be viewed at wileyonlinelibrary.com]

the traditional finite difference method (FD) in Figure 6. In FD method, the cost function gradient is simply determined by changing each design variable and once flow solving with that change. Since the same flow solvers and mesh perturbation techniques have been used for both FD and ALBM, the two curves in Figure 6 should be theoretically identical. The convergence criteria for the flow and adjoint solvers were 10^{-6} and 10^{-3}, respectively. Results indicate that these convergence criteria are sufficient to obtain highly consistent results for both methods. Therefore, the differences in two curves which have similar trend and different magnitude at some locations, probably, result from an inconsistency of the particular discretization used for the continuous adjoint equation.[43,44] However, the similar trend of general variation of present adjoint method with FDM indicate that the present method is reliable.

After 50 design cycle, the normalized cost function is reduced about 0.05 and conversancy is achieved. The comparison of pressure coefficient distribution, initial, target and optimal profiles are presented in Figure 7. As seen in Figure 7, very excellent adjustment for optimal and target shape is obtained. Though this optimization is performed in unsteady flow condition at time 1.5 but fast convergence and accurate shape recovery is obtained.

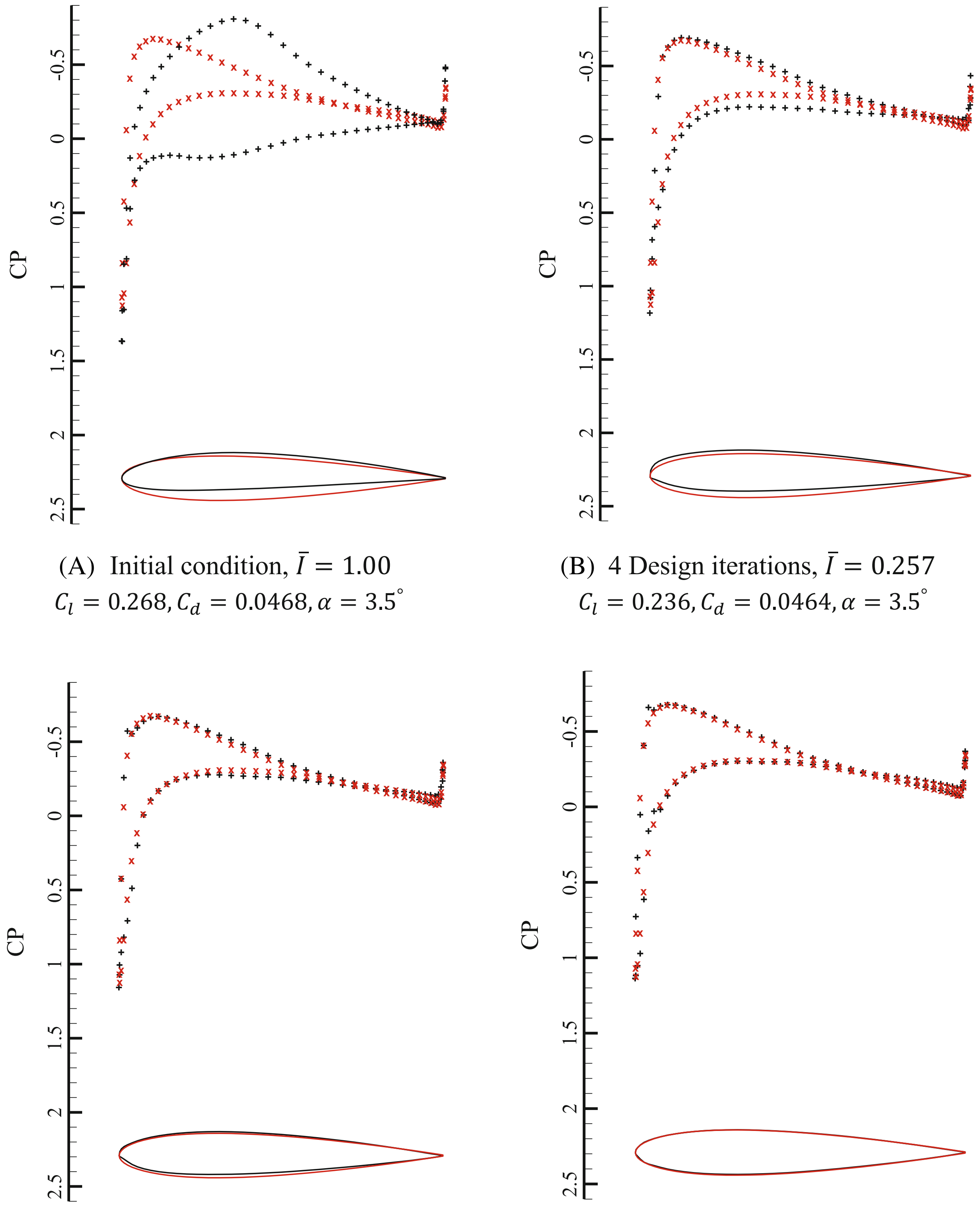
The history of the cost function and the gradient norm $|G| = \sum\left(\sqrt{\overline{\mathcal{G}}_x^2 + \overline{\mathcal{G}}_y^2}\right)$ versus design cyclesand the residual history of flow and adjoint solution are presented in Figures 8 and 9, respectively. The results indicate that new optimization approach based on the CF idea is accurate and correctly work.

The history of gradient vector and shape of airfoil during optimization procedure are shown in Figures 10 and 11, respectively. Figure 10 clearly show that gradient distribution is tend to zero at the end of design cycles.

Finally, the Mach number contour, grid and stream line of optimal airfoil is shown in Figure 12.

7.2 | Case II: Steady compressible laminar flow ($M_\infty = 0.8, Re_\infty = 400, \alpha = 2.31°$)

In this test case, an airfoil inverse design is presented in the compressible flow. The goal is to reach to the CAST10-2 transonic airfoil as the target shape from the ONERA-D airfoil as the initial shape at final time 4.5 where the steady state condition is occurred. The optimization problem is performed at Mach number of 0.8 and Reynolds Number 400 within 2.31° angle of attack. The total number of surface point is 81 nodes as the design variables. The cost function gradient versus design variables is shown in Figure 13 for first design cycle. Also, for the validation of derivatives, this values are compared with the traditional finite difference method (as a reference values) and is signed with "**o**" versus "+" for the present ALBM in Figure 13. It is obvious that accurate estimation of cost function gradient has been achieved by the presented algorithm. The error level that is apparent in Figure 13 is smaller to Figure 6 for the CASE I.



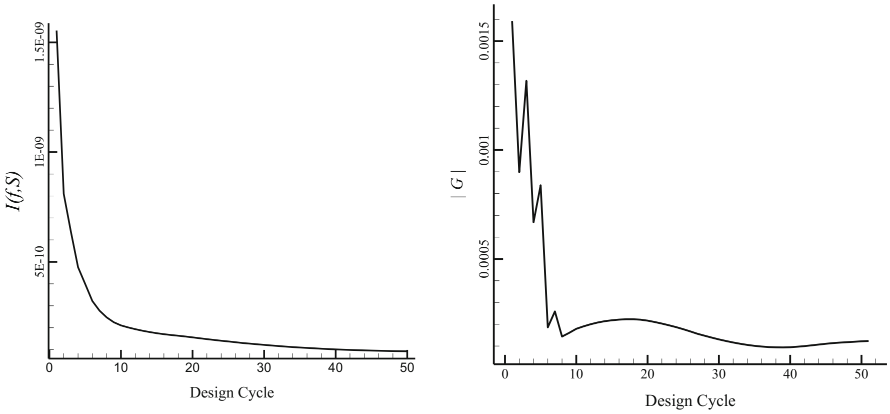
(A) Initial condition, $\bar{I} = 1.00$
$C_l = 0.268, C_d = 0.0468, \alpha = 3.5^{\circ}$

(B) 4 Design iterations, $\bar{I} = 0.257$
$C_l = 0.236, C_d = 0.0464, \alpha = 3.5^{\circ}$

(C) 10 Design iteration,$\bar{I} = 0.135$
$C_l = 0.186, C_d = 0.0503, \alpha = 3.5^{\circ}$

(D) 50 Design iterations, $\bar{I} = 0.05$
$C_l = 0.168, C_d = 0.0504, \alpha = 3.5^{\circ}$

FIGURE 7 Fixed alpha mode incompressible case inverse design, 81 design variables, $M_\infty = 0.25$, Re_∞=500, $\alpha = 3.5^{\circ}$, time = 1.5, (+) Initial Airfoil: NACA2410. (×) Target Cp: NACA0012, $C_l = 0.16$, $C_d = 0.0512$. [Colour figure can be viewed at wileyonlinelibrary.com]



(A) Cost function

(B) Norm of gradient

FIGURE 8 History of (A) Cost function variation. (B) Norm of gradient vector.

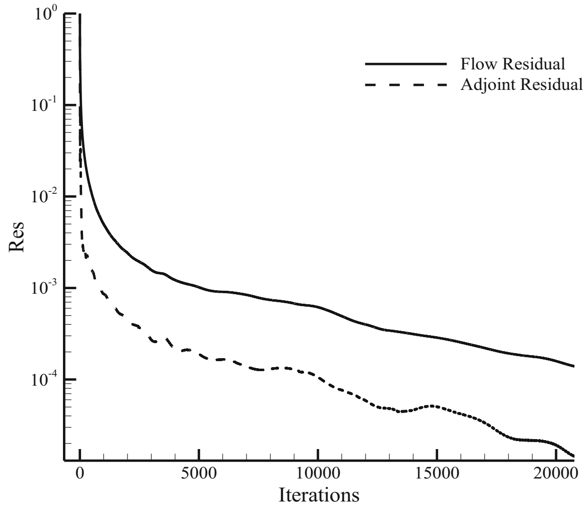


FIGURE 9 Flow and adjoint residual history.

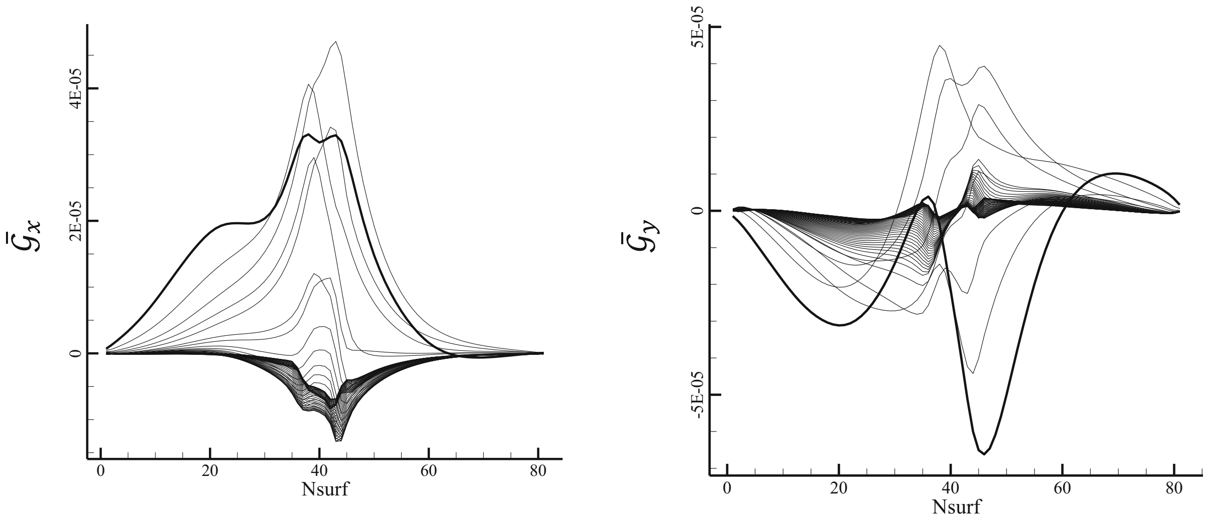


FIGURE 10 Variation of cost function gradient vector during optimization in Case I.

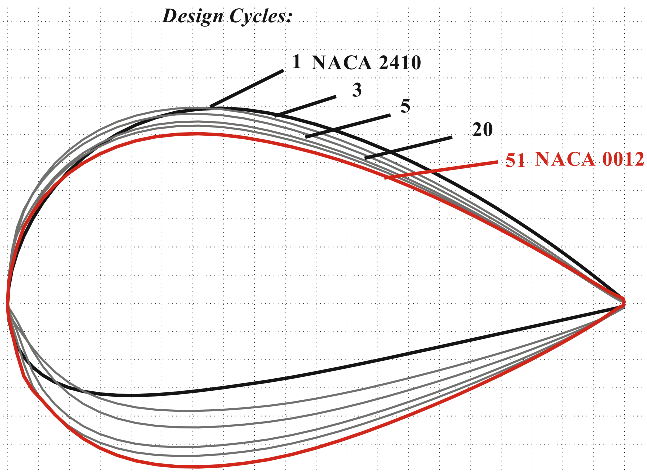


FIGURE 11 Variation of airfoil shape in optimization process in Case I. [Colour figure can be viewed at wileyonlinelibrary.com]

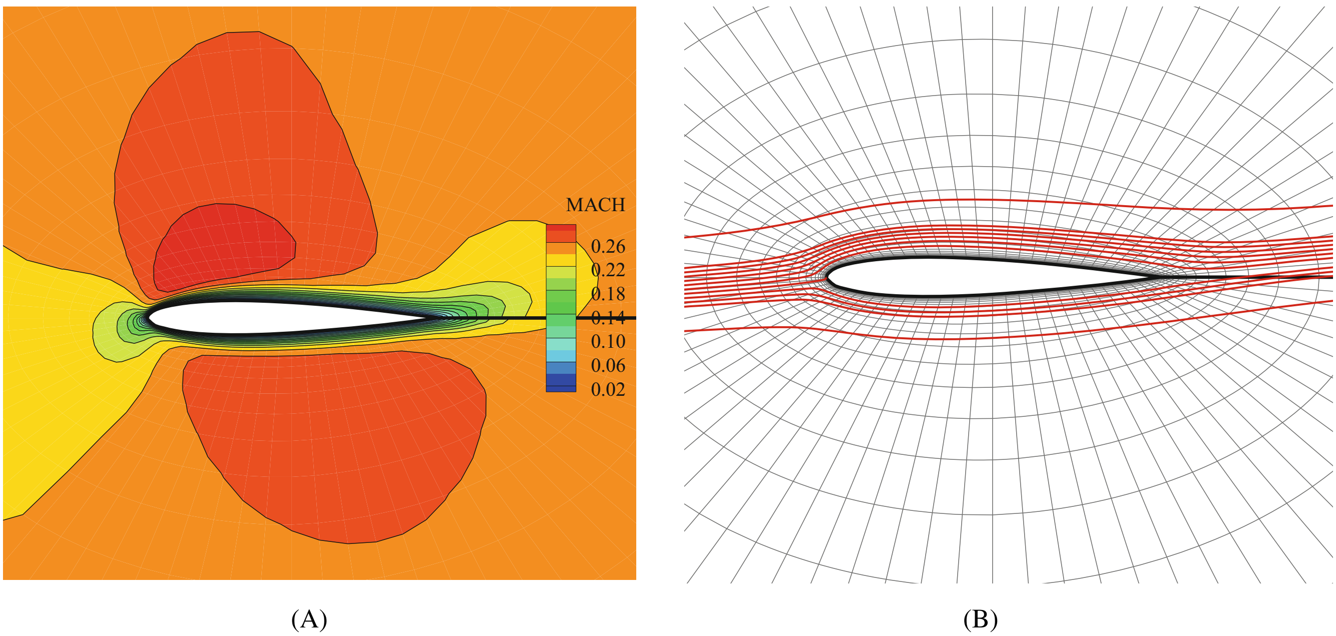


FIGURE 12 CASE I Optimal airfoil, A) Mach No. contour, B) O-type grid and flow stream lines. [Colour figure can be viewed at wileyonlinelibrary.com]

After about 75 design cycle the normalized cost function is reduced about 0.05 and convergence is achieved. The comparison of the pressure coefficient distribution, initial, target and optimal profiles are presented in Figure 14. As seen, excellent agreement for optimal and target shape is obtained at steady state conditions. In this problem, the optimization step size, λ, is changed linearly from 1.23 to 0.26 during design procedure.

The history of the cost function and gradient norm versus design cycles is shown in Figure 15. The results indicate that new developed optimization algorithm also accurately works at compressible flows. The flow and adjoint residual history in Figure 16 shows the convergence in this case is occurred. Also, the history of gradient vector and shape of airfoil during optimization procedure for this case are shown in Figures 17 and 18, respectively.

Finally, the Mach number contour, grid and stream line of optimal airfoil of this case are shown in Figure 19.

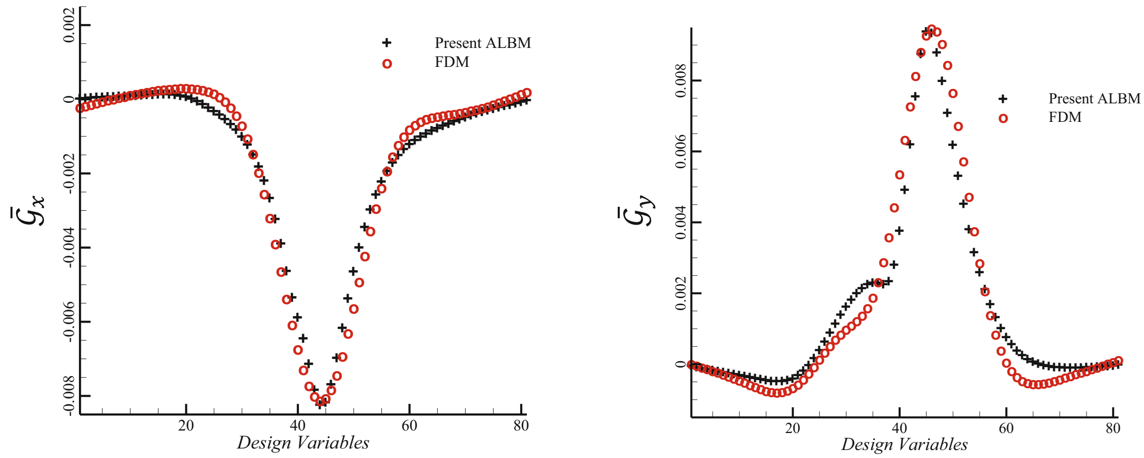
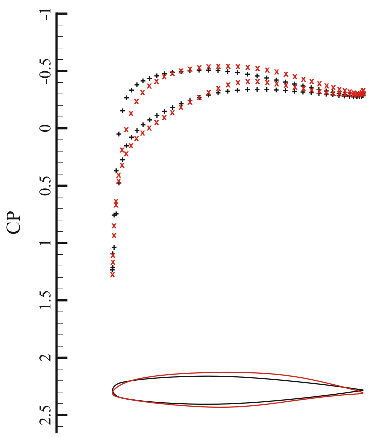
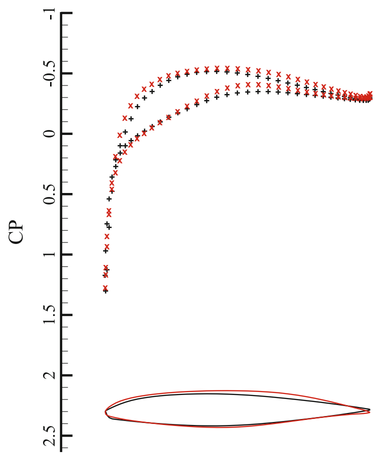


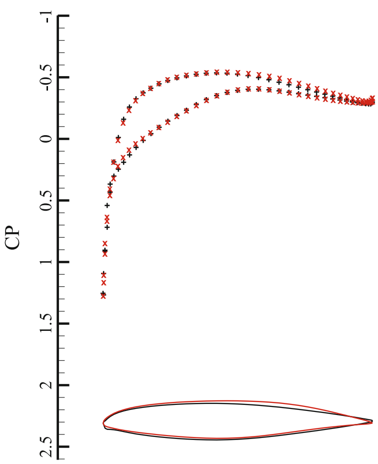
FIGURE 13 Distribution of smoothed gradients of the cost function $\left(\bar{\mathcal{G}}_x, \bar{\mathcal{G}}_y\right)$, comparison of present ALBM and FDM gradient in first design cycle. [Colour figure can be viewed at wileyonlinelibrary.com]



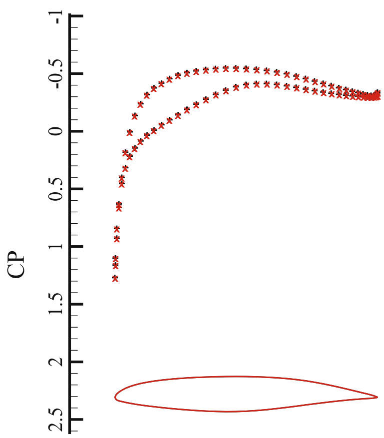
(A) Initial condition, $\bar{I} = 1.00$
$C_l = 0.157, C_d = 0.0776, \alpha = 2.31^{\circ}$



(B) 5 Design iterations, $\bar{I} = 0.577$
$C_l = 0.135, C_d = 0.07091, \alpha = 2.31^{\circ}$

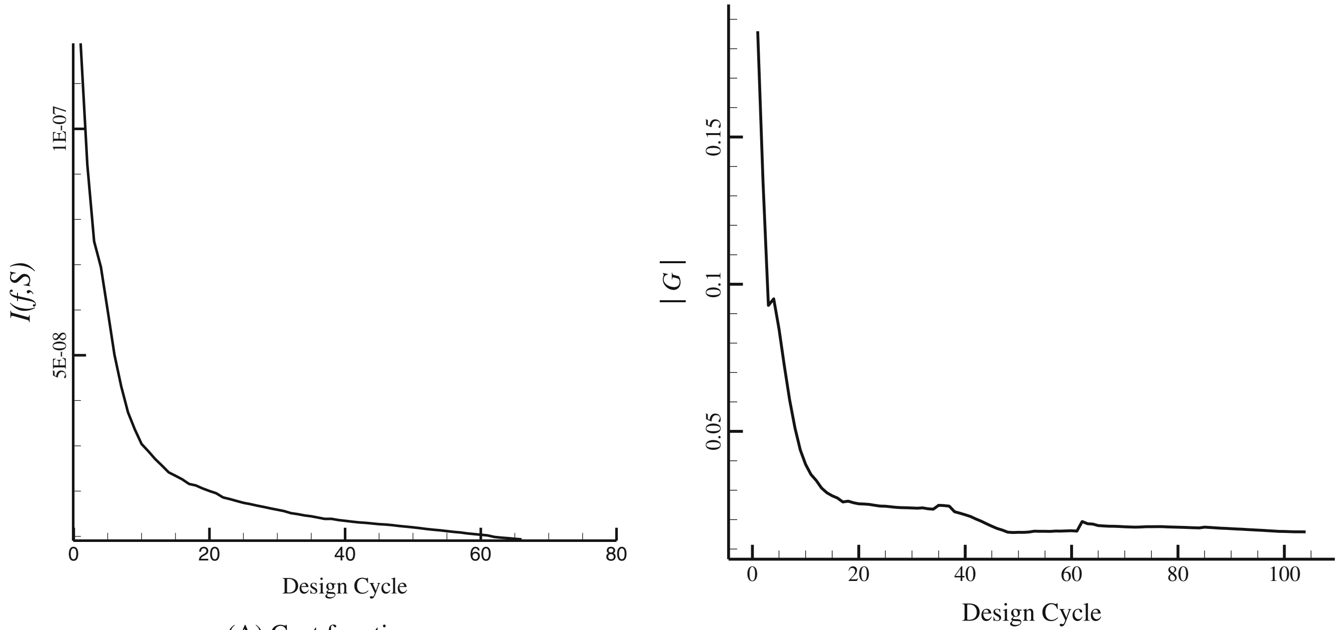


(C) 40 Design iteration, $\bar{I} = 0.112$
$C_l = 0.141, C_d = 0.0865, \alpha = 2.31^{\circ}$



(D) 75 Design iterations, $\bar{I} = 0.05$
$C_l = 0.146, C_d = 0.0857, \alpha = 2.31^{\circ}$

FIGURE 14 Fixed alpha mode compressible case inverse design, 81 design variables, $M_\infty = 0.8$, $\mathrm{Re}_\infty = 400$, $\alpha = 2.31°$, time = 4.5, (+) initial airfoil: ONERA-D. (×) Target Cp: CAST10-2, $C_l = 0.145$, $C_d = 0.0856$. [Colour figure can be viewed at wileyonlinelibrary.com]



(A) Cost function

(B) Norm of gradient

FIGURE 15 History of the cost function variation.

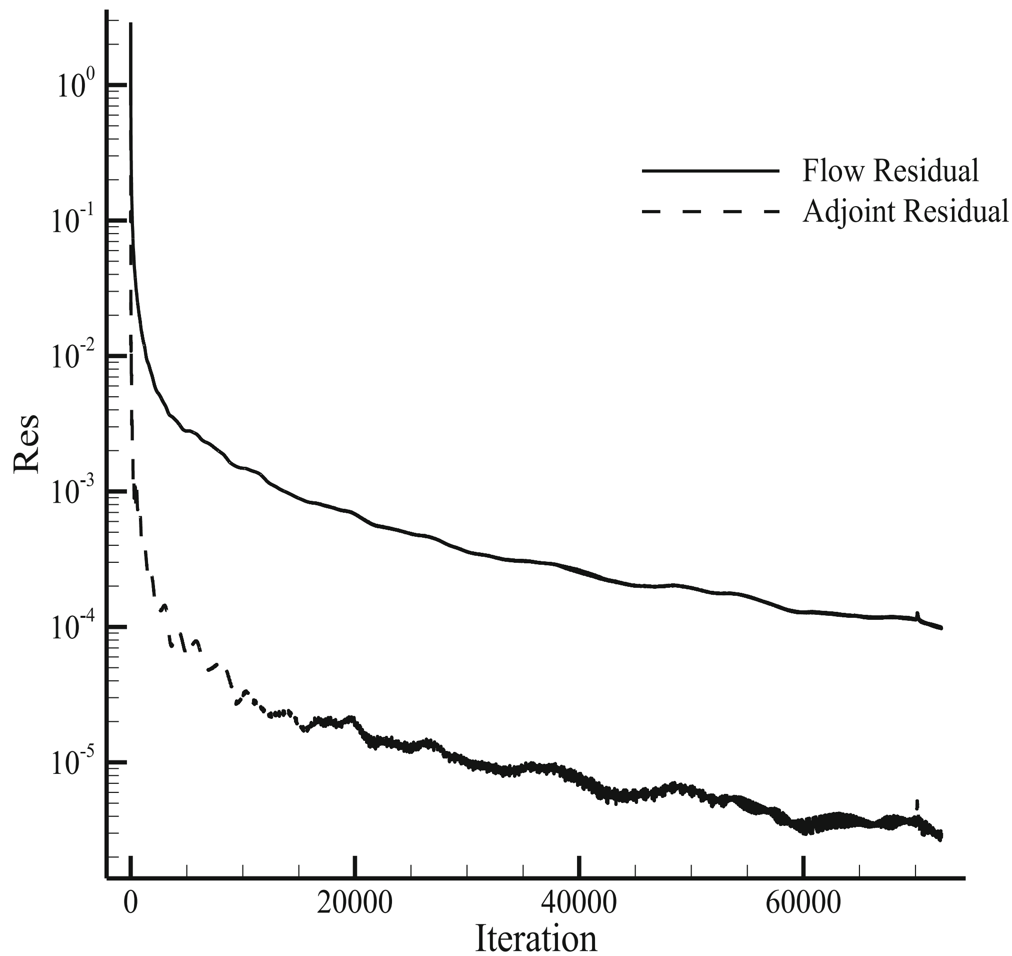


FIGURE 16 Flow and adjoint residual history.

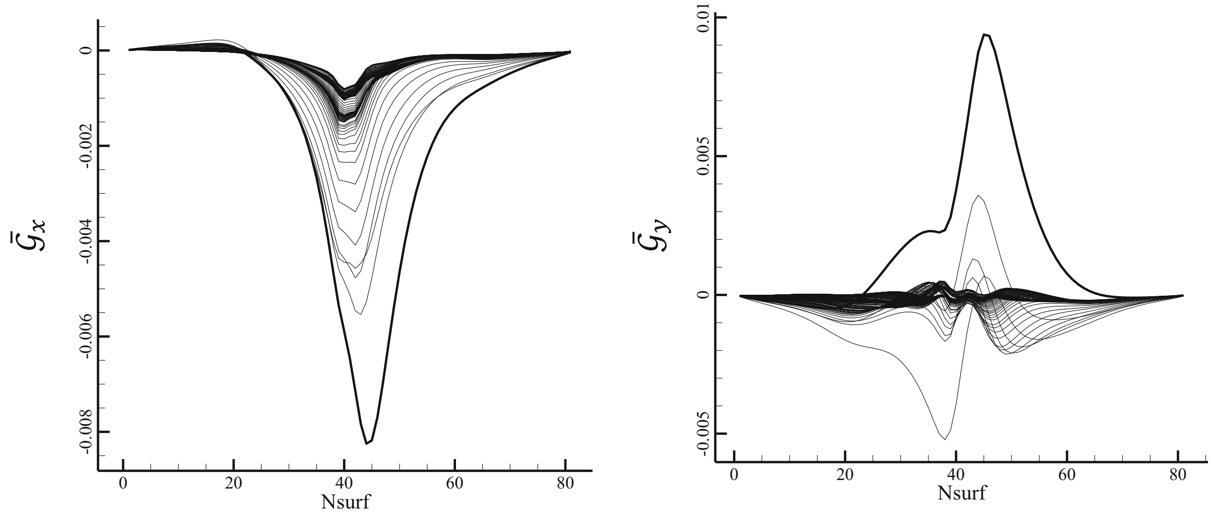


FIGURE 17 Variation of cost function gradient vector during optimization in CASE II.

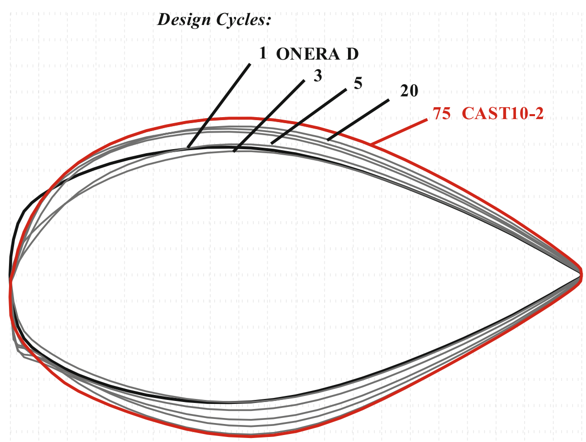


FIGURE 18 Variation of airfoil shape in optimization process in CASE II. [Colour figure can be viewed at wileyonlinelibrary.com]

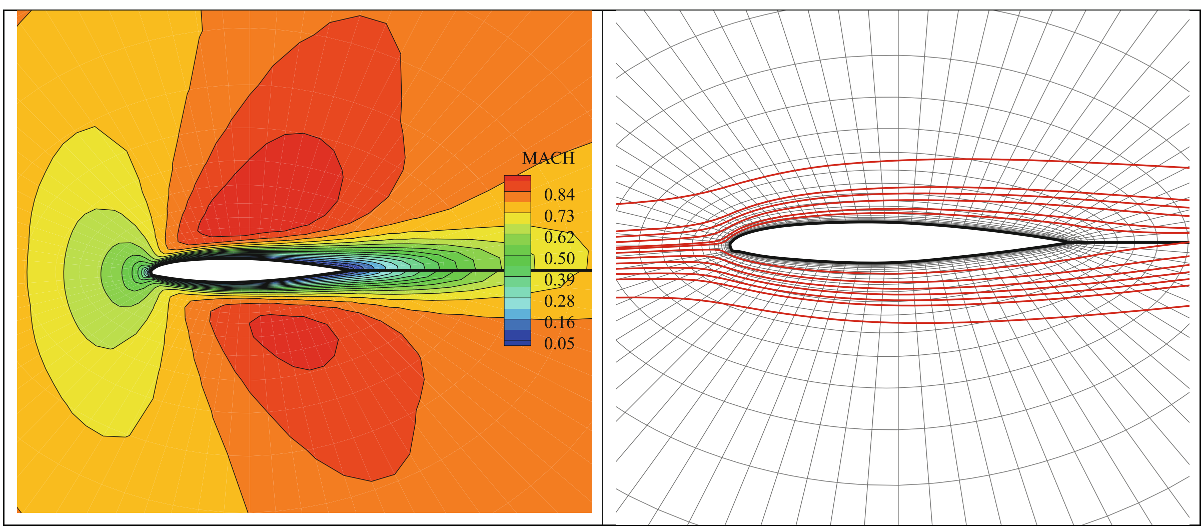


FIGURE 19 CASE II Optimal airfoil, (A) Mach No. contour, (B) O-type grid and flow stream lines. [Colour figure can be viewed at wileyonlinelibrary.com]

8 | CONCLUSION

In this study, a new approach of combination of the compressible LBM and adjoint method has been developed for the first time. The control theory based on the airfoil inverse design has been developed for the LBE in compressible flows. The adjoint finite volume lattice Boltzmann method (AFV-LBM) based on the circular function (CF) idea has been successfully derived for estimation of the cost function gradient vector which is suitable for optimization of both the compressible and incompressible viscous airfoil inverse design problems. The main advantage of the adjoint method based on the LBM versus NS or Euler equations is its simplicity for coding and adjoint derivation. However, ability of the LBM to cover simulation of high Knudsen number flows in rarefied conditions and advantages of the adjoint method make this approach interesting for future developments. In order to overcome singularity problem of boundary condition in the adjoint derivation, the new D2Q12L2 lattice has been presented in this paper with elimination of zero particle velocity from the lattice. In fact the flexibility of the CF scheme in new lattice generation has been used in this way.

In order to verification the developed optimization algorithm, two test case problem of airfoil inverse design have been studied. The first problem was the conversion of the NACA2410 to NACA0012 airfoil at the unsteady laminar incompressible flow. The second problem was the conversion of ONERA-D airfoil to transonic CAST10-2 at the viscous compressible steady flow. By evaluation the results of these two problems, capability of the proposed optimization algorithm has been examined for different flow conditions. The results indicate that new developed optimization algorithm accurately works at both incompressible and compressible flows. In the future work, we tend to deriving general equation of FV-ALBE for practical cost function in airfoil and wing design.

ACKNOWLEDGMENTS

I would like to express my very great appreciation to Dr. H. Farrokhfall for his valuable and constructive suggestions during the planning and development of this research work. His willingness to give his time so generously has been greatly appreciated.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article

ORCID

R. Kamali-Moghadam https://orcid.org/0000-0002-0780-7689

REFERENCES

1. Hu R, Jameson A, Wang Q. Adjoint-based aerodynamic optimization of supersonic biplane airfoils. *J Aircr*. 2012;49(3):802-814.
2. Kim S, Hosseini K, Leoviriyakit K, Jameson A. Enhancement of class of adjoint design methods via optimization of parameters. *AIAA J*. 2010;48(6):1072-1076.
3. Lee JD, Jameson A. NLF airfoil and wing design by adjoint method and automatic transition prediction. Paper presented at: 27th AIAA Applied Aerodynamics Conference; 2009, p. 3514.
4. Yamashita Y, Okamoto Y. Design optimization of synchronous reluctance motor for reducing iron loss and improving torque characteristics using topology optimization based on the level-set method. *IEEE Trans Magn*. 2020;56(3):1-4.
5. Lozano C. Entropy and adjoint methods. *J Sci Comput*. 2019;81(3):2447-2483.
6. Qian YH, Succi S, Orszag SA. Recent advances in lattice Boltzmann computing. *Annu Rev Comput Phys*. 1995;III:195-242.
7. Fei L, Luo KH, Li Q. Three-dimensional cascaded lattice Boltzmann method: improved implementation and consistent forcing scheme. *Phys Rev E*. 2018;97(5):053309.
8. Saadat MH, Bösch F, Karlin IV. Semi-Lagrangian lattice Boltzmann model for compressible flows on unstructured meshes. *Phys Rev E*. 2020;101(2):023311.
9. Sbragaglia MR, Benzi R, Biferale L, Succi S, Sugiyama K, Toschi F. Generalized lattice Boltzmann method with multirange pseudopotential. *Phys Rev E*. 2007;75(2):026702.
10. Li Q, He YL, Wang Y, Tang GH. Three-dimensional non-free-parameter lattice-Boltzmann model and its application to inviscid compressible flows. *Phys Lett A*. 2009;373(25):2101-2108.
11. Gharibi F, Ashrafizaadeh M. Simulation of high-viscosity-ratio multicomponent fluid flow using a pseudopotential model based on the nonorthogonal central-moments lattice Boltzmann method. *Phys Rev E*. 2020;101(4):043311.
12. Guo Z, Zhao TS. Lattice Boltzmann model for incompressible flows through porous media. *Phys Rev E*. 2002;66(3):036304.
13. Wang J, Zhang X, Bengough AG, Crawford JW. Domain-decomposition method for parallel lattice Boltzmann simulation of incompressible flow in porous media. *Phys Rev E*. 2005;72(1):016706.

14. Najafi MJ, Naghavi SM, Toghraie D. Numerical simulation of flow in hydro turbines channel to improve its efficiency by using of lattice Boltzmann method. *Phys A: Stat Mech Appl*. 2019;15(520):390-408.
15. Sun C, Hsu AT. Three-dimensional lattice Boltzmann model for compressible flows. *Phys Rev E*. 2003;68(1):016303.
16. Shouxin H, Guangwu Y, Weiping S. A lattice Boltzmann model for compressible perfect gas. *Acta Mech Sinica*. 1997;13(3):218-226.
17. Wang Y, He YL, Zhao TS, Tang GH, Tao WQ. Implicit-explicit finite-difference lattice Boltzmann method for compressible flows. *Int J Mod Phys C*. 2007;18(12):1961-1983.
18. Qu K, Shu C, Chew YT. Alternative method to construct equilibrium distribution functions in lattice-Boltzmann method simulation of inviscid compressible flows at high Mach number. *Phys Rev E*. 2007;75(3):036706.
19. Qu K, Shu C, Chew YT. Lattice Boltzmann and finite volume simulation of inviscid compressible flows with curved boundary. *Adv Appl Math Mech*. 2010;2(5):573-586.
20. Qu K, Shu C, Chew YT. Simulation of shock-wave propagation with finite volume lattice Boltzmann method. *Int J Mod Phys C*. 2007;18(4):447-454.
21. Qu K, Shu C, Cai J. Developing LBM-based flux solver and its applications in multi-dimension simulations. *Int J Mod Phys Conf Ser*. 2012;19:90-99.
22. Wilde D, Krämer A, Reith D, Foysi H. Semi-Lagrangian lattice Boltzmann method for compressible flows. *Phys Rev E*. 2020;101(5):053306.
23. Jameson A, Shankaran S, Martinelli L. Continuous adjoint method for unstructured grids. *AIAA J*. 2008;46(5):1226-1239.
24. Nadarajah SK, Jameson A. Optimum shape design for unsteady flows with time-accurate continuous and discrete adjoint method. *AIAA J*. 2007;45(7):1478-1491.
25. Jameson S, Jameson A. Adjoint formulations for topology, shape and discrete optimization. Paper presented at: 45th AIAA Aerospace Sciences Meeting and Exhibit; 2007, p. 55.
26. Kim S, Alonso JJ, Jameson A. Multi-element high-lift configuration design optimization using viscous continuous adjoint method. *J Aircr*. 2004;41(5):1082-1097.
27. Farrokhfal H, Pishevar AR. Aerodynamic shape optimization of hovering rotor blades using a coupled free wake–CFD and adjoint method. *Aerosp Sci Technol*. 2013;28(1):21-30.
28. Farrokhfal H, Pishevar AR. Optimization of airfoils for minimum pitching moment and compressibility drag coefficients. *J Aerosp Technol Manag*. 2014;6:395-406.
29. Jalali Khouzani H, Kamali MR. A novel approach of unsteady adjoint lattice Boltzmann method based on circular function scheme. *J Sci Comput*. 2020;85(2):1-29.
30. Cheylan I, Fritz G, Ricot D, Sagaut P. Shape optimization using the adjoint lattice Boltzmann method for aerodynamic applications. *AIAA J*. 2019;57(7):2758-2773.
31. Li X, Fang L, Peng Y. Airfoil design optimization based on lattice Boltzmann method and adjoint approach. *Appl Math Mech*. 2018;39(6):891-904.
32. Dugast F, Favennec Y, Josset C, Fan Y, Luo L. Topology optimization of thermal fluid flows with an adjoint lattice Boltzmann method. *J Comput Phys*. 2018;15(365):376-404.
33. Chen C, Yaji K, Yamada T, Izui K, Nishiwaki S. Local-in-time adjoint-based topology optimization of unsteady fluid flows using the lattice Boltzmann method. *Mech Eng J*. 2017;4(3):17-120.
34. Rokicki J. Adjoint lattice Boltzmann for topology optimization on multi-GPU architecture. *Comput Math Appl*. 2016;71(3):833-848.
35. Vergnault E, Sagaut P. An adjoint-based lattice Boltzmann method for noise control problems. *J Comput Phys*. 2014;1(276):39-61.
36. Hekmat MH, Mirzaei M. Extraction of macroscopic and microscopic adjoint concepts using a lattice Boltzmann method and discrete adjoint approach. *Phys Rev E*. 2015;91(1):013303.
37. Hekmat MH, Mirzaei M. Development of discrete adjoint approach based on the lattice Boltzmann method. *Adv Mech Eng*. 2014;22(6):230854.
38. Hekmat MH, Mirzaei M. Continuous and discrete adjoint approach based on lattice Boltzmann method in aerodynamic optimization part I: mathematical derivation of adjoint lattice Boltzmann equations. *Adv Appl Math Mech*. 2014;6(5):570-589.
39. Hekmat MH, Mirzaei M. A comparison of the continuous and discrete adjoint approach extended based on the standard lattice Boltzmann method in flow field inverse optimization problems. *Acta Mech*. 2016;227(4):1025-1050.
40. Jameson A. Aerodynamic design via control theory. *J Sci Comput*. 1988;3(3):233-260.
41. Drela M. XFOIL: an analysis and design system for low Reynolds number airfoils. *Low Reynolds number aerodynamics*. Springer; 1989:1-12.
42. Kun Q. Development of lattice Boltzmann method for compressible flows.
43. Reuther J. Aerodynamic shape optimization using control theory No. NAS 1.15:201064. 1996.
44. Kroll N, Schwamborn D, Becker K, Rieger H, Thiele F, eds. MEGADESIGN and MegaOpt-German Initiatives for Aerodynamic Simulation and Optimization in Aircraft Design: Results of the Closing Symposium of the MEGADESIGN and MegaOpt Projects, Braunschweig, Germany, May 23 and 24, 2007. Springer Science & Business Media; 2009.

How to cite this article: Jalali Khouzani H, Kamali-Moghadam R. Airfoil inverse design based on laminar compressible adjoint lattice Boltzmann method. *Int J Numer Meth Fluids*. 2023;95(8):1197-1219. doi: 10.1002/fld.5192